

07

QA & Testing Strategy

How TrustHome PH is tested: test levels and tools, what counts as evidence, status and severity definitions, the test-account matrix, the execution order for the next two weeks, and the release exit criteria. It is the companion to the Use Cases & Test Cases workbook (Rev 2).

Revision	Rev 2.0 — 1 October 2026
Supersedes	Test-approach parts of the QA Documentation; the "How to use" notes in the old workbook
Stack baseline	React + Firebase Spark (Auth, Firestore, Hosting) + Cloudinary Free + Vercel Hobby API + GitHub Actions. No paid plan, no credit card.
Audience	Prades (QA lead), every tester, Scrum report writer
Team	Andresa (Scrum Master) · Carillo · Laurente · Prades · Solpico

READ THIS FIRST

The workbook now holds 48 use cases and 202 test cases. As of 1 Oct, 9 have passed live, 27 have passed only in development, 4 are blocked, 160 are not yet run, and 2 are retired. The biggest QA risk is not missing tests. It is counting unit-test or emulator results as release evidence. This strategy separates the two, defines exactly what "Passed" means, and orders execution so the 23 cases unblocked by the 1 Oct rules deployment run first. The new API and private-document cases (TC-API, TC-DOC) run as soon as v1.4 is live. Release requires zero open Critical defects and every Critical case passed live.

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1. Quality goals

1. **Prove the trust claims.** Verified means reviewed, bookings never overlap, and private data stays private. These are the cases the panel will probe.
2. **Evidence over assertion.** A case is "Passed" only with a tester name, a date and evidence from the **deployed** system.
3. **Fix forward, safely.** Every defect fix ships with a regression test.

2. Test levels

Level	What	Tool	Where	Counts as release evidence?
Unit	Pure logic: validation, overlap, trust/fairness math, loan, admin-guard decisions	<code>node --test</code> , Python <code>unittest</code>	Local + CI	No: supports <i>Passed (Dev)</i>
Rules	Allowed/denied per actor per collection	<code>@firebase/rules-unit-testing</code> + Firestore emulator	Local + CI	No, but required before any rules deploy
API integration	Endpoints against emulators: auth, 401/403/409, concurrency	<code>node --test</code> in <code>api-server/</code> , <code>vercel dev</code>	Local + CI	No
Live functional	Workbook cases through the real UI	Browser, test accounts	<code>trusthome-ph.web.app</code>	Yes
Live security	Bypass the UI: direct SDK writes, raw HTTP, other users' tokens	Browser console snippets, <code>curl</code> , small scripts	Live project	Yes
Non-functional	Accessibility, performance, cross-browser, mobile	axe DevTools, Lighthouse, real phones	Live	Yes
Smoke / E2E	Renter-owner-admin journey end to end	TC-E2E-001 script	Live, after every deploy	Yes

Current automated baseline (1 Oct): 47 Node tests, 23 Python tests, 5 Firestore emulator rules tests, Functions-emulator overlap and concurrency tests, ESLint, production build, `npm audit` with 0 vulnerabilities.

3. Status definitions (use exactly these in the workbook)

Status	Meaning	Required fields
Passed	Executed on the deployed test project; actual = expected	Actual Result, Tester, Date, evidence link in Remarks
Passed (Dev)	Passed only by unit test, emulator or local run	Remarks: which test. Must be re-run live before release
Failed	Executed; actual ≠ expected	Actual Result + defect ID (F-xx) in Remarks

Status	Meaning	Required fields
Blocked	Cannot run yet	Remarks names the blocker (F-05, F-16, "v1.4 not deployed")
Not Run	Runnable, not yet executed	—
Retired	Feature removed; kept for traceability	Remarks: replacement case

WATCH OUT • Do not change a case to Passed because "the code looks right" or "the unit test passes". The progress report already warns against marking workbook cases complete based on unit tests alone. The Summary sheet's pass rate counts *Passed (Dev)* separately for this reason.

4. Evidence and defect logging

Evidence: a screenshot (with URL bar and signed-in account visible) or a console/HTTP log, saved to the team's shared QA folder as `TC-ID_YYYYMMDD.png`. Put the file name in Remarks.

Defect record (F-xx):

Field	Example
ID / title	F-17: Confirm button stays enabled after 409
Severity	Critical / High / Medium / Low (definitions below)
Steps to reproduce	Numbered, with account used and data
Expected vs actual	One line each
Evidence	File name
Linked TC / FR	TC-API-005 / FR-027
Owner / status	Laurente / Open → Fixed → Verified

Severity	Definition	Release rule
Critical	Security or privacy breach, data loss, double booking, cannot complete a core journey	Must be fixed and verified
High	Core feature wrong with no workaround; wrong trust or fairness output	Fix, or document a workaround approved by the team
Medium	Feature works with a workaround; confusing UX; non-critical accessibility	Fix if time allows; list in release notes
Low	Cosmetic	Backlog

5. Test accounts and data

Create these on the live project before 5 Oct. Keep the passwords in the team password manager, never in the workbook.

Account	Role / state	Owns	Used for
qa.admin@...	admin, active	One listing (admin-who-also-owns case)	Review, disputes, admin feed isolation
qa.owner1@...	user, active	2 rent listings (1 nightly, 1 monthly) + 1 sale listing	Intake, purpose, requests, confirm via API
qa.owner2@...	user, active	1 listing, no requests	Empty states, "other owner" denial tests
qa.renter1@... / qa.renter2@...	user, active	—	Competing requests, concurrency, disputes
qa.suspended@...	user, suspended	1 listing	Suspended-account screen, API 403, hidden listing
qa.noprofile@...	Auth user with no users doc	—	Admin guard fail-closed (TC-ADM-006)
Guest	signed out	—	Public visibility, direct-read denials

Data rules: synthetic only. Use the labelled placeholder documents from the seeder; never upload a real ID. Mark every QA listing title with [QA] so it can be cleaned up. Reset demo data on 13 Oct (TC-OPS-006).

6. Execution order (next two weeks)

Wave	When	Cases	Why first
1	2–6 Oct	23 cases unblocked by the 1 Oct rules deploy: TC-AUTH-010, TC-INTAKE-013, TC-VER-002/005, TC-BOOK-004/014/016, TC-RATE-002/005, TC-NOTIF-001/003/004/006/007/009, TC-MAP-005/007/008, TC-BANK-003, TC-ADM-007, TC-SEC-001/005/006	Rules are live now. These prove the security baseline
2	2–7 Oct	Oct-1 features: TC-PURP-001...006, TC-NOTIF-012...015; admin access TC-ADM-003...006	Fresh code, deployed, never tested live
3	7–8 Oct	v1.4: TC-API-001...010, TC-DOC-001...009, TC-OPS-001...003, TC-SEC-002/003, TC-VER-006, TC-NOTIF-002	Closes F-05, F-16, F-01
4	9 Oct	Re-run all Passed (Dev) cases live (27 cases)	Converts dev evidence to release evidence
5	9 Oct	Remaining Not Run by module: Booking → Disputes → Trust → Map → My Listings → Navigation/Settings → Bank → Release	Highest-risk modules first
6	12 Oct	TC-E2E-001 ×2, TC-UI-001/002/003, TC-HARD-001...003	Final confidence and accessibility

6.1 How to run security cases (bypassing the UI)

Direct Firestore write (TC-SEC-001/002 and every "direct write" case). Use a small Node script with the **client** SDK, not the Admin SDK, so the rules apply exactly as they do for a browser:

```
// qa/direct-write.mjs → QA_EMAIL=... QA_PASSWORD=... node qa/direct-write.mjs <bookingId>
import { initializeApp } from 'firebase/app';
import { getAuth, signInWithEmailAndPassword } from 'firebase/auth';
import { getFirestore, doc, updateDoc } from 'firebase/firestore';
const app = initializeApp({ /* same public config as the web app */ });
await signInWithEmailAndPassword(getAuth(app), process.env.QA_EMAIL, process.env.QA_PASSWORD);
try {
  await updateDoc(doc(getFirestore(app), 'bookings', process.argv[2]), { status: 'Confirmed' });
  console.log('FAIL: write was allowed');
} catch (e) { console.log('PASS:', e.code); } // expected: permission-denied
```

Copy and adapt it per case: change the account, the document path and the forbidden field.

Raw API call (TC-API-002/003):

```
curl -i -X POST https://<api>.vercel.app/api/bookings/<id>/confirm # expect 401
curl -i -X POST -H "Authorization: Bearer <reenter ID token>" https://<api>/api/bookings/<id>/confirm
# expect 403
```

Concurrency (TC-API-006): a Node script signs in the owner, then fires two `confirm` calls for two overlapping Pending bookings with `Promise.all`. Repeat 10 times on fresh seed pairs. Pass only if every run yields exactly one Confirmed and one 409.

7. Non-functional checks

Check	Tool	Target
Accessibility	axe DevTools on U1-U9 and A1-A3; keyboard-only run of Create Listing and Booking	0 serious/critical issues
Contrast	Token table in 05 §3	All text ≥ 4.5:1
Performance	Lighthouse mobile on Home, Browse, Listing detail	Performance ≥ 80; LCP < 2.5 s
Browsers	Chrome (Android), Safari (iOS), Chrome + Firefox + Edge (desktop)	Core journey works
Viewports	360, 390, 768, 1280 px	No horizontal scroll on user pages
Quota	Firebase Usage after a full test day	< 50% of daily Spark limits

8. Entry and exit criteria

Entry (start of a test wave): build deployed and tagged; rules commit recorded; test accounts ready; backup taken if the wave includes migrations.

Exit (release v1.5.0):

- ☐ 0 open Critical defects; every High defect fixed or has an approved workaround.
- ☐ Every **Critical** test case is **Passed** (live), including TC-API-005/006, TC-DOC-001/004/009, TC-SEC-001/002, TC-OPS-001.
- ☐ No case left as *Passed (Dev)* for a Critical or High requirement.
- ☐ TC-E2E-001 passed twice on the final tagged build.

- ☐ Security release gate (04 §9) fully checked.
- ☐ Workbook Summary sheet exported and attached to the final Scrum report.

9. Metrics to report weekly

Metric	Source	Why
Executed / total, by module	Summary sheet	Coverage progress
Live pass rate (Passed ÷ Executed)	Summary sheet	Quality, without dev-only inflation
Passed (Dev) remaining	Summary sheet	Evidence debt
Open defects by severity	Defect log	Release readiness
Blocked count + blockers	Summary sheet + Remarks	What the critical path is waiting on

10. Workbook maintenance rules

- Edit only the yellow columns (Status, Actual Result, Tester, Date). Use the dropdown.
- New case: next number in its prefix. Fill UC, FR, Module (must match a Summary module name, or the "Unassigned module check" goes above 0), and set Origin = New.
- A changed requirement updates the case's Expected Result and sets Origin = Updated, with the reason in Remarks.
- Never delete a case. Set it to Retired and name its replacement.
- After each wave, copy the Summary numbers into the Scrum report and the progress report.